INCH-POUND

MIL-S-83121A(USAF)
31 January 1989
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MILITARY SPECIFICATION

SNOWPLOW, TRUCK MOUNTING, UNDERBOY MXX-513/S32U

This specification is approved for use within the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of hydraulically operated, underboy type snowplow, designated MXX-513/S32U.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

Federal

QQ-C-320	Chromium Plating (Electrodeposited)
QQ-S-781	Steel Strapping, Flat
TT-V-121	Varnish, Spar, Water-Resisting
UU-T-81	Tags, Shipping and Stock
PPP-B-601	Box, Wood, Cleated-Plywood
MIL-B-121	Barrier Material, Greaseproofed, Waterproofed, Flexible

Beneficial comments (recommendation, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMEDTB, Robins AFB, GA 31096-5609 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMC N/A FSC 3825

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PPP-B-621	Box, Wood, Nailed and Lock-Corner
*PPP-P-40	Packaging and Packing of Hand Tools
PPP-T-60	Tape, Pressure-Sensitive Adhesive, Waterproof, for Packaging

Military

MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted
MIL-P-116	Preservation, Methods of
MIL-P-514	Plates, Identification, Instruction and Marking, Blank
MIL-F-3541	Fittings, Lubrication
MIL-C-3774	Crates, Wood, Open, 12,000- and 16,000-Pound Capacity
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirement for
MIL-H-13531	Hose, Rubber and Hose Assembly, Rubber (Hydraulic, Flexible)
MIL-V-13811	Varnish, Waterproofing, Electrical, Ignition
MIL-T-83118	Trucks, Snow Removal Equipment A/S32U-35 and A/S32U-36 (Heavy Duty)
*MIL-C-83282	Hydraulic Fluid, Fire Resistant, Synthetic, Hydrocarbon Base, Aircraft

STANDARDS

Federal

FED-STD-151	Metals, Test Methods
*FED-STD-595	Colors

Military

MIL-STD-12	Abbreviations for use on Drawings and in Technical-Type Publications
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-808	Finishes, Protective, and Codes, for Finishing Schemes for Ground and Ground Support Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-1186	Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

(Copies of specifications, standards, and other Governments documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer).

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Society of Automotive Engineers, Incorporated

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SAE Standards and Recommended Practices

J739	Grader Cutting Edge
J740	Countersunk Square Holes for Cutting Edges and End Bits

(Application for copies should be addressed to the Society of Automotive Engineers Incorporated, Two Pennsylvania Plaza, New York, New York 10001).

United States of America Standards

B18.9	Plow Bolts

(Application for copies should be addressed to the United States of America Standards Institute, 10 East 40th Street, New York, New York 10016).

*2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for associated detail specifications, specification sheets or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Preproduction. This specification makes provisions for preproduction testing.

3.2 Components The snowplow shall consist of the following components.

Item Name	See Requirement
Frame	3.6.1
Moldboard	3.6.2
Hydraulic components	3.6.3
Rotating and adjustment mechanism	3.6.4

3.3 Materials. Materials shall be as specified herein. Materials not definitely specified shall be of the best commercial quality and entirely suitable for the purpose. Materials shall be free of all defects and imperfections that might affect the serviceability of the finished product.

3.4 Design and construction. The snowplow shall be designed and constructed as an underbody scraper assembly for mounting on and functioning with snow removal equipment trucks conforming to MIL-T-83118. All parts necessary for installation of the equipment on the truck and for efficient and satisfactory operation of the snowplow when connected to the truck controls shall be furnished. All parts of the snowplow shall be of such size, material, and strength as to withstand the full output power of the truck in all gear ratios and under all operating conditions specified herein. In all positions of the

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moldboard, the snowplow shall operate without blade chatter, weaving, rocking, or other actions not under complete control of the operator.

3.4.1 Road clearance. Road clearance shall be not less than 10 inches with the moldboard in the raised and horizontal position.

3.4.2 Reliability. The snowplow shall be designed and constructed as specified herein and to successfully complete the tests specified in 4.4.2 with no failure preventing satisfactory performance. The snowplow shall be so designed and constructed that parts will not work loose in service. It shall be built to withstand the strains, jars, vibrations, and other conditions incident to shipping, storage, and service.

3.4.3 Maintainability. The snowplow shall be designed and constructed as specified herein and to provide the following:

a. Minimum number of parts consistent with reliability and performance specified herein.

b. Minimum amount of training and time necessary for assembly, disassembly, location of trouble sources, and maintenance including servicing. Where practical, parts and components shall be located or positioned for rapid and simple inspection and recognition of excessive wear or potential failure.

c. Permit adjustments, servicing, replacement of parts and components, and other maintenance with minimum disturbance to other equipment parts or components. Parts and components shall be located for ample and raped access unless performance or reliability will be appreciable degraded by the accessible location. In performing maintenance, if engineering reasoning or data determines that physical or visual interference between items cannot be avoided, the item predicted to require the most maintenance shall be located for best accessibility.

d. Permit maintenance with general-purpose tools and equipment normally available commercially. Use of special-purpose tools and equipment shall be subject to approval by the procuring activity.

e. Minimum number of tools required for maintenance by design practices such as reducing the variety of bolt head sizes to the practicable minimum.

f. Foolproofness. - Where improper installation of an item could cause malfunction of the item or the equipment in which it is installed, an unsymmetrical mounting means shall be provided where practical. That mounting shall allow the item to be installed only in its proper operating position. If an unsymmetrical mounting means is not practical, the item shall be so mounted that its proper operating position can be readily and visible determined by service maintenance personnel.

3.5 Performance. The snowplow, when installed on an A/S32U-35 snow removal equipment truck in accordance with MIL-T-83118, ballasted to an operating weight of 46,000 pounds (including this snowplow) shall withstand the following conditions without detrimental effect to subsequent operation.

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3.5.1 Operation. The snowplow shall operate satisfactorily in ice, hard-packed snow, gravel, clay, or loam for not less than 8 hours at the full output power of the truck. The snowplow shall withstand a minimum evenly distributed force of 25,000 pounds applied horizontally across the bottom third of the plow blade when in the straight across (bulldoze) position.

3.5.2 Environmental conditions. The snowplow shall withstand the following conditions without detrimental effect to subsequent operation:

a. Operating temperatures ranging from -40° to +125°F.

b. Storage temperatures ranging from -80° to +160°F.

3.6 Details of components.

3.6.1 Frame. the frame shall be fabricated or structural steel of sufficient strength to support the moldboard and accessories and to withstand severe plowing stresses. Plowing stresses shall be transmitted to the truck chassis by transferring horizontal forces from a heavy-duty swing circle through a drawbar to the forward section of the chassis and vertical forces through heavy-duty shock springs to the chassis. Ample clearance between the plow and the truck shall be provided for the specified plow movements.

3.6.2 Moldboard. The moldboard constructed of abrasion-resistant steel shall attach to the heavy-duty swing circle and be correctly curved for efficient displacement of ice and hard-packed snow. It shall be reinforced along its entire length by substantial structural steel. The moldboard shall be drilled to permit installation of a four-piece blade cutting edge with hole spacings in accordance with SAE Standard J739.

3.6.2.1 Blade cutting edge. The blade cutting edge shall conform to the SAE Standard J739. The cutting edge shall be constructed of abrasion-resistant steel, shall be removable in four sections, and shall be curved to fit the moldboard. The cutting edge shall be not less than 1/2 inch thick and 6 inches high. The bolt holes shall be in accordance with SAE Standard J740. Number 3 head plow bolts grade 8 in accordance with USA Standard B18.9 with nut and lockwasher shall be used. Standard mill tolerances for material of this class will be acceptable. The cutting edge shall have a Rockwell hardness of C25 to C35. The two outside corners of the cutting edge shall be rounded using a 3-inch radius.

3.6.3 Hydraulic components. The hydraulic components shall consist of rams, hosed, piping, fittings, and all other parts essential to the practical and efficient operation of the snowplow. These components shall be compatible with the control system specified in MIL-T-83118. A minimum number of fittings, joints, and connections shall be used to prevent undue back pressure, clogging, and leakage. All parts shall be insulated from detrimental vibration and shock. Rams that actuate the underbody blade from horizontal (travel position to vertical (plowing) position shall be capable of exerting a force sufficient to lift the ballasted truck.

3.6.3.1 Flexible hose and hydraulic tubing. Hose for hydraulic lines shall conform to MIL-H-13531, type II, class A. The hydraulic tubing and flexible hose shall be of adequate size to permit free flow of hydraulic fluid conform-

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ing to MIL-C-83282 at the temperatures specified in 3.5.2.

3.6.3.2 Hydraulic rams. Pistons rods and interior cylinder surfaces of all hydraulic rams shall be plated with chromium in accordance with QQ-C-320, type II.

3.6.3.3 Gages. Two hydraulic pressure gages indicating nominal operating pressure midway on the gage shall be furnished and mounted near the hydraulic control valves to indicate pressure being exerted by the blade cutting edge.

3.6.4 Rotation provisions. Not less than six hinges shall be equally spaced along the length of the moldboard to permit rotation of the moldboard in a 90° arc from the vertical to the horizontal. Rotation of the moldboard shall be accomplished by at least two double-acting hydraulic rams.

3.6.4.1 Moldboard adjustment. The moldboard shall be adjustable in a horizontal plane through not less than 80° centered laterally (40° fore and aft) to the longitudinal axis of the snow removal truck to permit displacement of snow to either side. A minimum of six stop positions shall be provided in the swing circle to adjust the moldboard cutting angle. A minimum of two hydraulically-actuated mechanical locks shall maintain the moldboard in any of the stop positions. Power adjustment of the moldboard shall be required.

3.7 Lubrication. All moving parts shall be provided with suitable means of lubrication and shall operate satisfactorily when lubricated with standard military lubricants as specified in MIL-M-8090.

3.7.1 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541 and shall be located in accessible, protected positions. Extended fittings shall be provided to lubricate parts or assemblies that are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.7.2 Lubrication chart. A lubrication chart shall be provided directing attention to all lubrication fittings and shall specify the range and grade of lubricant required for critical temperatures. The chart shall be permanently attached in a protected, conspicuous location.

3.7.3 Initial service. The snowplow shall be completely serviced prior to delivery.

3.8 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-STD-130 shall govern the manufacturer's part numbers and changes thereto.

3.9 Dimensions. the moldboard shall be not less than 5/8 inch in nominal thickness, 12 feet long, and 13 inches high as measured over the blade cutting edge of the moldboard.

3.10 Finishes and protective coatings. The snowplow parts normally painted shall be painted in accordance with the manufacturer's commercial practice. The color shall be Forest Green, Color Number 24052 of FED-STD-595 conforming to MIL-STD-208.

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3.11 Markings

3.11.1 Data plates. Transportation data plates conforming to MIL-P-514 shall be provided. All marking plates, placards, charts, instruction plates, et cetera, shall conform to MIL-P-514. The size and shape shall be consistent with information required thereon. Abbreviations shall be in accordance with * MIL-STD-121.

3.12 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.13 Government-loaned property. When specified in the contract (see 6.2), the Government will loan one A/S32U-35 snow removal equipment truck conforming to MIL-T-83118 to the contractor upon his request.

3.14 Workmanship. Workmanship shall be of the highest grade throughout and in accordance with good commercial practice for this type of equipment.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of tests. The inspection and testing of the snowplow shall be classified as follows:

a. Preproduction testing . . . . . See 4.4.

b. Acceptance tests . . . . . See 4.5.

4.3 Test conditions

4.3.1 Mounting. The snowplow shall be mounted on an A/S32U-35 snow removal equipment truck conforming to MIL-T-83118 for the tests specified in 4.5.3 * and 4.6.5. The truck shall be ballasted to an operating weight of 46,000 pounds (including this snowplow). For all other tests, a fixture, in lieu of the truck, may be used to simulate operating characteristics. The fixture

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shall duplicate the truck mounting, controls, pressures, and hydraulic action required to operate all components.

4.3.2 Preparation for test. Prior to testing, the snowplow shall be serviced with lubricants in accordance with the instructions of the manufacturer.

4.4 Preproduction testing (see 6.2).

4.4.1 Test sample. One snowplow shall be subjected to the tests specified in 4.4.3.

4.4.2 Test report. After completion of the preproduction tests, a test report shall be prepared in accordance with MIL-STD-831 and three complete copies of the report furnished the procuring activity.

4.4.2.1 Reliability and maintainability information. The following information shall be included as an appendix to the test report:

a. All failures, servicing, adjustments, maintenance, and irregular functioning shall be identified by accumulated operating time, cycles, miles, or position in the test procedure, as appropriate. The conditions at the time of the events identified shall be recorded.

b. A summary of the engineering analysis and of any tests conducted to determine assignable causes for any failure or irregular functioning.

c. A summary of the engineering analysis leading to any corrections made to design, construction, quality control, or other procedures, or leading to any corrections to be made to production items or proposed to be made. The summary shall also include an analysis of the predicted effectiveness of such corrections.

d. Clock time and man-hours required for each maintenance and servicing action taken during the tests. A brief description of the experience and qualifications of the personnel taking such actions shall be included.

e. Test activity or contractor comments on item features or requirements that, if modified, should improve the item.

f. Test activity or contractor comments on use or maintenance conditions to be avoided or cultivated to increase the reliability or useful life of the item.

g. Any of the above information that is already included in the test report body need not be repeated in the information required by this paragraph, but clear reference to the location of the data shall be included.

4.4.3 Preproduction tests. The preproduction tests shall consist of all tests specified under 4.6.

4.5 Acceptance tests. The acceptance tests shall consist of the following:

a. Individual test.

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b. Sampling test.

4.5.1 Individual test. Each snowplow shall be subjected to an examination of product test as described under 4.6.

4.5.2 Sampling test

4.5.2.1 Periodic sampling. One snowplow shall be selected at random from each 25 or fraction thereof produced and subjected to the test specified in 4.6.2, except it shall be reversed only 10 times.

4.5.2.1.1 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, all of the tests shall be repeated.

4.5.2.1.2 Individual tests may continue. For operational reasons, individual tests may be continued pending the investigation of a test failure. Final acceptance of the items on hand or produced later shall not be made until it is determined that all items meet all the requirements of the specification.

4.5.3 Defects in items already-accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all the defects likely to be found and the method of correcting them.

4.6 Test methods

4.6.1 Examination of product. The snowplow shall be visually examined for compliance with the requirements of this specification with regard to completeness of assembly, presence of all components and accessories, quality of workmanship and materials, dimensions, hydraulic component leaks, marking, mounting suitability and simplicity, and the functional operation of all components, assemblies, accessories, and controls.

4.6.2 Mounting and functioning test. The snowplow shall be mounted on the truck or fixture as specified in 4.3.1 and all dimensions, fitting locations, and operating clearances shall be checked. All operable components shall be operated to determine that they function satisfactorily. The snowplow shall be reversed a minimum of 100 times (from maximum cutting angle in one direction to maximum angle in the opposite direction).

4.6.3 Strength test. When mounted in accordance with 4.3.1, an evenly distributed 25,000-pound force shall be applied to the snowplow as specified in 3.5.1. There shall be no impairment of function when subjected to this force nor permanent deformation or any parts after removal of this force.

4.6.4 Hardness test. The blade cutting edge shall be subjected to a hardness test in accordance with FED-STD-151 to determine compliance with the requirements of 3.6.2.1.

4.6.5 Operational test. The truck-mounted snowplow shall be operated to

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determine that it functions satisfactorily as specified herein. Actual plowing operations shall be accomplished in either ice, hard-packed snow, gravel, clay, or loam for a minimum of 100 hours to determine that the various components are of adequate strength and function properly. The maximum force shall be applied to the blade cutting edge. No evidence of hydraulic leaking or component failure shall occur.

4.6.5.1 Servicing. Only the following types of servicing will be permitted during the operational test:

a. Truck chassis servicing as recommended by the truck manufacturer.

b. Underbody snowplow:

(1) Daily lubrication or every 8 hours of operation.

(2) Tightening of bolts and fittings every 8 hours of operation.

(3) Rotation and replacement of blade cutting edge as required.

(4) Such other servicing as recommended and contained in service manuals for this type equipment and for the times specified in such manuals.

4.6.6 Environmental tests. The snowplow mounted in accordance with 4.3.1 shall be subjected to the following environmental tests in accordance with MIL-STD-810 for storage from -80° to +160°F and operator from -40° to +125°F. After stabilizing at operating temperature, 100 cycles of the following operations shall be performed:

a. Rotate the moldboard from a full down to a full up position and vice versa.

b. Rotate the moldboard in the horizontal plane from the extreme left position to the extreme right position and vice versa.

c. Engage the hydraulic lockpins at each stop position.

d. Raise and lower the moldboard in the vertical plane.

e. Apply a vertical force to the moldboard sufficient to lift the ballasted truck.

During of after the tests, failure of the rotating mechanism to smoothly rotate from one cast position to another, any evidence of hydraulic leakage or component failure, or inability to accomplish the required performance shall be cause for rejection.

4.6.6.1 High temperature. The snowplow shall be subjected to high temperature storage and operation in accordance with method 501, procedure I. After stabilizing at operating temperature, the snowplow shall be subjected to the test specified in 4.6.6.

4.6.6.2 Low temperature. The snowplow shall be subjected to low temperature

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storage and operation in accordance with method 502, procedure I. The storage temperature shall be maintained for 48 hours. After stabilizing at operating temperature, the snowplow shall be subjected to the test specified in 4.6.6.

4.6.7 Maintainability evaluation. The snowplow shall be inspected and evaluated from the standpoint of ease of maintenance, servicing, and operation. Particular attention shall be given maintenance with a minimum number of common tools, servicing, and operation by personnel wearing heavy gloves, and provisions made to prevent accumulation of dirt, snow, ice, et cetera, that may hinder servicing and operation.

4.7 Inspection of preparation for delivery. Preservation, packaging, packing, and marking shall be inspected to verify conformance to the requirements of section 5 herein.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified (se 6.2.

5.1.1 Level A

5.1.1.1 Disassembly. Disassembly shall be the minimum necessary to reduce cubage and safeguard parts known to be subject to damage, pilferage, and loss. Bolts, nuts, washers, and pins shall be placed in one of the mating parts and secured to prevent their loss.

5.1.1.2 Matchmarking. When necessary to facilitate reassembly, parts that are removed and mating parts shall be matchmarked with lusterless white enamel, overcoated with clear spar varnish conforming to TT-V-121. Small parts shall have matchmarking identification on waterproof, cloth shipping tags conforming to UU-T-81, type A, attached to mating parts.

5.1.1.3 Unpainted surfaces. Unpainted metal surfaces requiring the application of preservation in accordance with MIL-P-116, such as tie rods, adjusting rods, scarifiers, clevises, bolts and nuts, grease fittings, hose coupling and similar parts, shall be coated with type P-1 preservative. Unpainted metal information plates shall be coated with electrical ignition waterproofing varnish conforming to MIL-V-13811.

5.1.1.4 Machined surfaces. Unpainted exterior machined surfaces of component parts shall be coated with type P-2 or P-6 preservative. The coated surfaces shall be wrapped or covered with greaseproof barrier material conforming to MIL-B-121, secured in place waterproof, pressure-sensitive adhesive tape conforming to PPP-T-60, type III, class 1.

5.1.1.5 Hydraulic systems. The hydraulic fluid supply lines shall be flushed with the approved hydraulic fluid. Wherever practical, pistons shall be fully retracted and any exposed surfaces of the piston rods shall be coated with type P-6 preservative and spirally wrapped with greaseproof barrier material conforming to MIL-B-121, secured in place with waterproof, pressure-sensitive adhesive tape conforming to PPP-T-60, type III, class 1. Open ends of hose shall be sealed with tape conforming to PPP-T-60, type III, class 1.

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5.1.1.6 Operating and control levers. Unpainted control levers shall be coated with type P-1 preservative.

5.1.1.7 Extra cutting edges. Unpainted surfaces of extra cutting edges shall be coated with type P-1 preservative.

5.1.1.8 Maintenance tools. Maintenance tools shall be preserved and packaged in accordance with PPP-P-40.

5.1.1.9 Technical publications. Technical publications shall be packaged in accordance with MIL-P-116, method 1C1-1.

5.1.1.10 Intermediate packaging. Component parts removed from each snowplow and repair parts shall be Intermediate packaged in nailed wood boxes conforming to PPP-621, class 2, style optional or in wood cleated-plywood boxes conforming to PPP-B-601, overseas type, style optional.

5.1.2 Level C. The snowplow, component parts, tools, and publications shall be preserved and packaged in accordance with the manufacturer's standard practice.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The snowplow shall be packed in exterior containers conforming to MIL-C-104, or MIL-C-3774. Packaged and unpackaged components shall be placed and secured with flat steel strapping conforming to QQ-S-781 within the shipping container to prevent movement or damage to the components or the snowplow.

5.2.2 Level B. The snowplow shall be packed as specified for Level A, except the ends, sides, and top assemblies shall be omitted and the snow removal unit and component parts shall be shipped on the skid assembly of the container specification selected.

5.2.3 Level C. The snowplow, packaged and unpackaged components parts, tools, and publications shall be packed in compliance with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carriers rules applicable to the code of transportation.

5.2.3.1 Securing procedures. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186.

5.3 Marking. In addition to any special marking required, packages, shipping containers, and the snowplow shall be marked in accordance with MIL-STD-129. The nomenclature shall be:

Snowplow, Truck Mounted, Underbody MXK-513/S32U.

6. NOTES

6.1 Intended use. The MXK-513/S32U underbody-type snowplow is intended for permanent mounting on the A/S32U-35 or A/S32U-36 snow removal equipment truck for use in the removal of ice and hard-packed snow from airbase runways and taxiways and for maintenance grading of dirt and gravel roads.

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6.2 Ordering data. Procurement documents should specify the following:

a. That this snowplow shall only be procured as part of the A/S32U-35 or A/S32U-36 snow removal equipment truck in accordance with MIL-T-83118 (see 6.2 of MIL-T-83118).

b. Applicable levels of preservation, packing, and packing (see 5.1 and 5.2).

6.3 Subject term (key word) listing.

Snowplow
Truck Mounted
Underbody

6.4 Changes from previous issue. The margins of this specification are marked with asterisks (or vertical lines) to indicate where changes (additions, was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Air Force - 99

Preparing activity:
Air Force - 84

(Project 3825-F185)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-83121A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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(TO DETACH THIS FORM, CUT ALONG THIS LINE.)